



Weaning from CPB

Flow and drainage are reduced as deliberately as they were raised — every pre-wean condition must be confirmed first.

1

Physiologic Readiness For CPB Separation

2

Ventilations On

3

Surgeon: "Come Down To 3/4 Flow"

- A. **First** — fill the heart. Arterial waveform moves from flat to pulsatile; CVP rises 0 → 6.
- B. **Second** — decrease flow slowly to 3/4.
- C. **Third** — observe MAP.
- D. **Communication**

Vent on / off?

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Continue To Decrease Flow

Continue to decrease flow under the surgeon's direction until target baseline CVP is reached.

5

Surgeon: "Come Off Bypass"

Constant observation
Continuously observe the patient's hemodynamics post-CPB. Be ready for emergent re-initiation.

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Emergent Re-initiation

PATIENT PARAMETERS:

- ☐ Temperature — ~36–37°C (surgeon & procedure dependent)
- ☐ ECG — sinus rhythm, +/- pacing
- ☐ ABG — Hb/Hct, electrolytes (K⁺, Ca²⁺, Mg²⁺)
- ☐ Volume

"Are you happy with your ventilation?"

"Coming down to 3/4 flow"

FLOW REDUCTION REFERENCE

Venous open	CVP	Arterial waveform	Flow	Line pressure
100%	0	_____	Full	180
Decrease in MAP — maintain flow + ↑ filling (CVP) Increase in MAP — ↓ flow + maintain filling (CVP)				
75%	6		3/4	150

"I'm at 3/4 flow"

Remember
Maintain closed loop communication throughout.

"Coming off bypass venous is clamped arterial clamped off bypass"

- ☐ Close manifold
- ☐ Pause CO₂
- ☐ Close purge line
- ☐ Stop CPB timer
- ☐ Stop reperfusion timer
- ☐ Volatile off — Sevo, Iso

6

Drain (Or Sump) Venous

Venous cannula(s) out

7

Surgeon: "Give One Hundred"

Protamine 1 mg per 100 units heparin.

 **IMPORTANT**

Inform anesthesia of the total heparin dose, including prime on bypass.

Adverse Effects

Constant observation

Continuously observe the patient's hemodynamics post-CPB. Be ready for emergent re-initiation.

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Emergent Re-initiation

8

Surgeon: "Drain Arterial Line"

Shunt open

9

Charting

"Draining venous"

"I have 400mls in reservoir"

"Give one hundred, is your line clear?"

"Line clear"

"Giving one hundred"

"Hundred in, line clamped"

ADVERSE EFFECTS

- Hypotension
- Anaphylactoid reactions
- Pulmonary vasoconstriction

"Draining arterial line"

CENTRIFUGAL PUMP

- ☐ **RPMs zero**
- ☐ Remove arterial clamp
- ☐ RAP mode on
- ☐ Open auto clamp

- Volumes on pump